

Gauge Theory For Nonexperts

Fundamental to gauge theory is the idea of a local symmetry. To illustrate precisely what a local symmetry is, we look to familiar symmetries.

We use the analogy of converting currencies: We have different currencies that are associated with certain countries and definite methods for translating between them.

We have strictly defined methods for transferring between these different currencies regardless of the locations. We can also string together conversion rates for more complex currency exchange.

Notice that there is a definite exchange rate for any two locations and also notice that this exchange rate is somewhat arbitrary. Each currency is also 'arbitrary', we could also change the 'currency itself.'

E.g One 1 bill is 1/100 of a 100 bill, so the currency can be scaled up or down. This scaling affects the exchange rates by the same factors.

Now suppose a currency at every possible position (every point in space). This means that each location has a local currency.

If we allow currencies to vary across locations: There must be an exchange rate between any two arbitrary locations. For this, we use the concept of parallel transport.

Becoming more abstract, take a line, we also take a location which can vary at any point along the line, the values of these currencies on the line form "fibers". In this scenario, fibers are positive and real.

Allowing for locality creates fibers at every possible location.

Once we have fibers we need to know how to relate different fibers. In transferring between them, we convert between currencies, but in some sense we have the same underlying 'value'.

Parallel transport shows us about potential equivalencies in this value between fibers

We introduce the parallel transport operator which moves between two locations $P_{x(0)} \rightarrow P_{x(t)}$ (maps the fiber from $x(0)$ to the fiber over $x(t)$). This may result in something like: $y \rightarrow 20y$.

We have an arbitrary choice as to how to define our parallel transport operators. This gives us a continuous curve which seems to describe the same amount of value.

In this sense, we call parallel a generalization of 'constant' because these all have 'constant values' irrespective of their base.

In calculus we say that the derivative of a constant value is zero, so we want to introduce a new derivative which has 0 value for this parallelism.

In gauge theory we use a parallel section to capture this idea. A parallel section exists iff the covariant derivative vanishes. We show this by saying:

$f'(t) = -A(x(t))y(t)$ (Derivative A depends on $x(t)$ and multiplies $y(t)$ we call A a connection. A is a "matrix-valued 1-form")

By a section, we mean that a value is selected above each point in a 'base space'. e.g the surface of the earth for countries.

This parallelism is all with respect to the connection A (wrt exchange rate parallel transport)

A connection and a parallel transport operator specify the same thing. We take $y(0)$ and transport it to all other points of space.

We define $y(t) := P_{x(0) \rightarrow x(t)} y(0)$ and differentiating this gives the earlier $f'(t)$ form. A connection is the fundamental object of gauge theory.

Different connections specify different parallel transport operators.

A connection: A way of specifying movement between fibers.

Symmetry: The second aspect of gauge theory is symmetry. This related to the fact that we can vary the base symmetry without varying the fundamental value. This leads to the idea of gauge transformations:

- Can change base currency at every point (gauge freedom)

Ex: $1(1\text{dollarbill}) = .01(100\text{dollarbill})$ so $y \rightarrow .01y$

This base currency can vary location from location and that is a gauge transformation. The act of changing the base unit is a gauge transformation and it occurs at every location.

How does parallelism change in response to dynamism? - Notion of parallel should adjust "accordingly"

If you are moving through the US you are in a case of degenerate parallelism and you will notice that it collapses into being constant. Here we have $A = 0$!

In a different setting where we change currencies between two locations, the way that parallelism looks fundamentally changes. $A \neq 0$ here.

Gauge theorists care about connections in a modular way. We have to consider the connections which are gauge equivalent. We can say that these two scenarios are equivalent to the level of choice of gauge transformation.

Group of gauge elements forms a group: $g(x) = \text{rescale each currency at } x$.

$A(x) \rightarrow g(x)A(x)g^{-1}(x) + g(x)\frac{d}{dx}g^{-1}(x)$ we see that g and g inverse will cancel but since A becomes matrix valued we can move anywhere in the larger dimensional fiber, making g and g inverse matrix valued, changing the nature of multiplication.

This formula will give us an A that will make our two fibers parallel.

For fiber = $\mathbb{R}^+$, $g(x) = e^{r(x)} = A(x) + e^{r(x)} \frac{d}{dx} e^{r(x)} = A(x) - \frac{d}{dx} r(x)$

Old: $y' \rightarrow -Ay$.

New: $y' \rightarrow -(A - \frac{d}{dx} r(x))y = -Ay - \frac{d}{dx} r(x)y$

This example demonstrates how the change of gauge acts on connections and how it changes the notion of parallelism, although it is a simplified analogy.

"Non-abelian symmetries" are an idea which is fundamental to gauge theory. To illustrate local symmetries we discussed the abelian case:

$G = \mathbb{R}^+$ (positive group of rescaling). This group is abelian because the order of operations doesn't matter e.g $A \times B = B \times A$.

Non-abelian groups require special consideration in this sense. For an example we look at the group of rotations in 3d ($SO(3)$) "Special orthogonal group of order 3"

If we rotate something along the y axis and then rotate it along the x axis, but this is not the same as rotating along the x axis and then the y axis.

In the non-abelian city this kind of thing makes things nonlinear. This makes analysis difficult.

Why gauge theory? Gauge theory has been successfully applied in two primary areas: Particle physics– Particles can be derived from gauge symmetries.

Photons, gluons, and W,Z, bosons (mediate forces). If we look at these particles, they are essentially the components of a connection. These connection objects occur in condensed matter physics through Churn-Simons theory as well as superconductivity via Landau-Ginsberg theory.

Mathematics: - Differential geometry (fiber bundles) - Low dimensional topology + partial differential equations (yang-mills, anti self-dual, seibers-witten)